Md. Harun-Ur-Rashid

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ResearchGate Google Scholar GitHub

I am currently working as a Software Engineer in Bangladesh, concurrently serving as a Machine Learning mentor for AI/ML enthusiasts. Previously, I held a position as a Machine Learning Developer at Renwend Software Company, a Co-Founder, and an Instructor in my startup. In my research endeavors, I have led many freelance research projects with many international students, encompassing disease image classification and detection, the development of low-resource company name clearance search solutions, face matching systems, etc. These initiatives demonstrate my dedication to education and advancing research, industrial work in various domains, from deep learning to computer vision to natural language processing.

Employment History

2025 – Present

Software Engineer (Contract, On-site), Upay (UCB Fintech Company Limited), Bangladesh

- Built and maintained MFS systems using Django REST Framework, improving request handling efficiency.

2021 – 2025

Software Engineer (Full-time, On-site), Giga Tech Limited, Dhaka, Bangladesh

- Built and maintained face verification and recognition systems using CNN, Flask, and REST APIs for eKYC platforms.
- Improved OCR and Transliteration microservices performance by **30%**.
- Re-engineered the NID verification workflow into a stateless architecture.
- Led the migration to a distributed RPA architecture, reducing verification time by **40%**.
- Integrated with EC, CBS, AD, and Porichoy.
- Contributed to eKYC deployments for BRAC, MTBL, Basic Bank, IFIC, TAP, bKash, FSIBL.

2024 – 2025

Software Engineer (Machine Learning) (Contract, Remote), Syntech Solution Ltd., Dhaka, Bangladesh

- Built a microservice-based company name clearance search system using DRF, Whisper, PyTorch, Elasticsearch, and Word2Vec.
- Improved search accuracy by combining NLP and speech models.
- Containerized services using Docker and deployed to an Ubuntu server.
- Ensured consistent dev/prod environments for smooth deployments.
- Built a data analysis report system using Flask, pandas, and matplotlib.

Education

2015–2020 **B.Sc. in Computer Science and Engineering, Daffodil International University (DIU)**
Thesis Title: *Automatic Tag Prediction of Poems Using Bidirectional LSTM* [Link]
Result: CGPA 2.64 out of 4.00

Projects

- **Face-Based Employee Attendance System** *A system that automatically marks attendance by recognizing employees' faces. It uses deep learning for face detection and recognition, making the whole process faster and contactless.* Technologies: Python, Django, Flask, REST API, PostgreSQL
- **English and Bangla Fake News Detection System** *A tool designed for social media platforms to automatically detect and flag fake news in both English and Bangla. It uses deep learning to analyze text and classify whether the news is real or fake.* Technologies: Python, Flask, Selenium, PostgreSQL
- **Healthcare Management System (HMS)** *A complete web-based hospital management application where patients, doctors, and admins can manage appointments, medical records, billing, and reports. The system supports role-based access and has a clean, user-friendly interface.* Technologies: Python, Django, Flask, REST API, PostgreSQL
- **Nurse Finder System** *A platform that helps patients connect with nearby qualified nurses. Nurses can manage their profiles and availability, while patients can search by location or specialty and communicate securely through built-in messaging.* Technologies: Python, Django, REST API, PostgreSQL

Research Publications

Journal Articles

1. M. Raihan et al., "A deep learning and machine learning approach to predict neonatal death in the context of são paulo," *International Journal of Public Health Science (IJPHS)*, vol. 13, pp. 179–190, 2024. DOI: 10.11591/ijphs.v13i1.22577.

Conference Proceedings

1. M. K. Hosain, Harun-Ur-Rashid, T. B. Taher, and M. M. Rahman, "Genre recognition of artworks using convolutional neural network," in *2020 23rd International Conference on Computer and Information Technology (ICCIT)*, 2020, pp. 1–5. DOI: 10.1109/ICCIT51783.2020.9392688.

Skills

Languages	English, Bengali.
Coding	Python, Java, R.
Databases	MySQL, Oracle, PostgreSQL.
Web Dev	JavaScript, Django, Flask, FastAPI, Selenium.
Machine Learning	Supervised Learning (Regression, Classification), Unsupervised Learning (Clustering), Deep Learning (ANN, CNN, RNN, LSTM, transfer learning, VGG), ensemble learning (bagging, boosting, stacking), NLP (data preprocessing, word embedded, transformer), computer vision (CNN, RCNN, UNet).
Misc.	Microsoft Office(Excel), LaTeX, Overleaf
Operating systems.	Linux, Windows
Version Control	Git, GitHub

Skills (continued)

Tools & Libraries	Scikit-learn, NumPy, Pandas, Keras, Matplotlib, Anaconda, VSCode, Google Colab, Postman
Other Skills	Docker, Kubernetes, Jenkins

Achievements

Virtual

- **Robi Datathon 3.0, Dhaka, Bangladesh**: Top 20

Physical

- **NASA Space Apps Challenge- 2019, Dhaka, Bangladesh.** : 1st Runner up
- **International Business Competition-2019, Organized by Hult Prize of Daffodil International University, Dhaka, Bangladesh**: 1st Runner up (on Campus Round)

Professional Training and Certifications

Virtual

- **Artificial Intelligent and Machine Learning**: FTFL-Joint training by LICT, Dhaka University and Giga Tech Limited.
- **Deep Learning with Pytorch: Zero to GANs**: Jovian.ml

Talks

Virtual

- **Industrial Machine Learning project using ML & Real-Life Data processing**: UIU Robotics & United International University, Dhaka, Bangladesh.
- **Industrial Project On Machine Learning**: Career Development Club, CSE, UAP & University of Asia Pacific - UAP, Dhaka, Bangladesh.
- **Interactive Session on Machine Learning and Data Science** : Department of Computer Science and Engineering, IUBAT, Dhaka, Bangladesh.
- **Hands-on Machine Learning with Real-life Application From Scratch**: International Seminar and Workshop on Robotics, Machine Learning, Artificial Intelligence & IoT Tangail, Bangladesh
- **Python Programming Masterclass**: STEM CLUB Bangladesh, Dhaka, Bangladesh